School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Marginal PMF, Correlation, Covariance, Joint Gaussian Random Variables

Name: Samridhi Mehrotra

Enrollment No: AU2440033

Email: samridhi.m@ahduni.edu.in

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Overview

This lecture covers:

- Marginal PMF (definition, computation, example)
- Correlation and its limitations
- Covariance (definition, properties, interpretation)
- Pearson's correlation coefficient
- Joint Gaussian random variables

Definitions and Notation

Joint PMF

For discrete random variables X and Y:

$$p_{XY}(x, y) = P(X = x, Y = y)$$

Marginal PMFs

$$p_X(x) = \sum_y p_{XY}(x, y), \quad p_Y(y) = \sum_x p_{XY}(x, y)$$

Correlation (Raw Product Moment)

$$R_{XY} = E[XY]$$

Covariance

$$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$$

Equivalent form:

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$$

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Pearson Correlation Coefficient

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

$$-1 \le \rho_{XY} \le 1$$

Joint Gaussian Distribution

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp \Bigg(-\frac{1}{2(1-\rho^2)} \Bigg[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho \left(\frac{x-\mu_X}{\sigma_X}\right) \left(\frac{y-\mu_Y}{\sigma_Y}\right)$$

Marginal PMF

Key Idea: Marginal PMF is obtained by summing over the other variable.

$$p_X(x_i) = \sum_j p_{ij}, \quad p_Y(y_j) = \sum_i p_{ij}$$

$$\sum_i p_X(x_i) = 1, \quad \sum_j p_Y(y_j) = 1$$

Correlation

- Measures $E[XY]$
- Depends on scale
- Computed about origin
- $R_{XY} = 0$ implies orthogonality (not independence)

Covariance

Interpretation

- Positive: same direction
- Negative: opposite direction
- Zero: no linear relation

Properties

- Symmetry:
$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$
- Variance:
$$\text{Cov}(X, X) = \text{Var}(X)$$
- Linearity:
$$\text{Cov}(aX + b, cY + d) = ac\,\text{Cov}(X, Y)$$
- Independence:
$$X \perp Y \Rightarrow \text{Cov}(X, Y) = 0$$

Important:
$$\text{Cov}(X, Y) = 0 \not\Rightarrow X \perp Y$$

Pearson Coefficient

- Unit-free
- Measures strength of linear relationship

$$\rho > 0 \Rightarrow \text{positive correlation}$$
$$\rho < 0 \Rightarrow \text{negative correlation}$$
$$\rho = 0 \Rightarrow \text{no linear correlation}$$

Covariance Derivation

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

$$= E[XY - X\mu_Y - \mu_X Y + \mu_X\mu_Y]$$

$$= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y$$

$$= E[XY] - \mu_X\mu_Y$$

$$= E[XY] - E[X]E[Y]$$

Worked Example 1 (Marginal PMF)

Joint PMF table:

	$y = 0$	$y = 1$
$x = 0$	$\frac{1}{8}$	$\frac{1}{4}$
$x = 1$	$\frac{3}{8}$	$\frac{1}{4}$

Marginals:

$$p_X(0) = \frac{3}{8}, \quad p_X(1) = \frac{5}{8}$$

$$p_Y(0) = \frac{1}{2}, \quad p_Y(1) = \frac{1}{2}$$

Worked Example 2 (Covariance)

Given:

$$Y = -2X + 3$$

$$\mathrm{Cov}(X, Y) = -2\,\mathrm{Var}(X)$$

Worked Example 3

Given:

$$f_{XY}(x, y) = \frac{xy}{96}, \quad 0 < x < 4,\ 1 < y < 5$$

Results:

$$E[X] = \frac{8}{3}, \quad E[Y] = \frac{31}{9}$$

$$\mathrm{Var}(X) = \frac{8}{9}, \quad \mathrm{Var}(Y) = \frac{92}{81}$$

$$\mathrm{Cov}(X, Y) = 0$$

$$\rho_{XY} = 0$$

Summary

- Marginal PMF:

$$p_X(x) = \sum_y p_{XY}(x, y), \quad p_Y(y) = \sum_x p_{XY}(x, y)$$

- Correlation:

$$R_{XY} = E[XY]$$

- Covariance:

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

- Pearson coefficient:

$$\rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

- Key Result:

$$\mathrm{Cov}(X, Y) = 0 \nRightarrow X \perp Y$$